

Semester Test 1

Semestertoets 1

FSK116

16 March 2011 / 16 Maart 2011

Surname and initials <i>Van en voorletters</i>	<u>MEMO</u>	Student number <i>Studentenommer</i>	_____
Signature <i>Handtekening</i>	_____		
Time <i>Tyd</i>	2½ hours <i>2½ ure</i>	Max. Total <i>Maks. Totaal</i>	85 Marks Max <i>85 punte Maks</i>
Internal Examiner: <i>Interne Eksaminator</i>	Dr J M Nel	External Examiner <i>Eksterne Eksaminator</i>	Mr R Q Odendaal

• Answer all questions. • Clearly show all calculations. Do not just write down the answer. • Clearly state all assumptions, rules and laws used. • Where applicable draw the free-body diagrams. • Demonstrate clearly that you understand the physics that you use to answer the question. This means that a numerically correct answer will not necessarily be awarded full marks. • Programmable calculators may NOT be used. • No textbooks allowed. • A formula sheet is attached. • All test and examination regulations of the University of Pretoria are applicable.	• <i>Beantwoord alle vrae.</i> • <i>Wys al jou berekeninge. Moenie net 'n antwoord neerskryf nie.</i> • <i>Stel duidelik alle aannames, reëls en wette wat u gebruik.</i> • <i>Waar van toepassing, teken die vryliggaamsdiagram (of kragte diagramme).</i> • <i>Wys dat jy die fisika wat jy gebruik om die vraag te beantwoord verstaan. Dit beteken dat volpunte sal nie noodwendig vir 'n numeries-korrekte antwoord gegee word nie.</i> • <i>Programmeerbare sakrekenaars MAG NIE gebruik word nie.</i> • <i>Geen handboeke word toegelaat nie.</i> • <i>'n Formuleblad is aangeheg.</i> • <i>Alle toets- en eksamenregulasies van die Universiteit van Pretoria is van toepassing.</i>
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Circle: Circumference $= 2\pi r =$
omtrek

$$\text{Area} = \pi r^2$$

Sphere: Area $= 4\pi r^2$

$$\text{Volume} = \frac{4}{3}\pi r^3$$

Cylinder / Silinder

$$\text{Area} = 2(\pi r^2) + 2\pi rh$$

$$\text{Volume} = \pi r^2 h$$

Trapezium: Area $= \frac{1}{2}(a+b)h$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\sin \alpha \pm \sin \beta = 2 \sin \frac{1}{2}(\alpha \pm \beta) \cos \frac{1}{2}(\alpha \mp \beta)$$

$$\vec{a} \cdot \vec{b} = ab \cos \phi$$

$$\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z$$

$$\vec{a} \times \vec{b} = (a_y b_z - b_y a_z) \hat{i} \\ + (a_z b_x - b_z a_x) \hat{j} \\ + (a_x b_y - b_x a_y) \hat{k}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$T_c = T_K - 273.15^\circ$$

$$T_F = \frac{9}{5} T_C + 32$$

$$\Delta L = L \alpha \Delta T$$

$$\Delta V = V \beta \Delta T$$

$$Q = C(T_f - T_i)$$

$$Q = cm(T_f - T_i)$$

$$Q = Lm$$

$$W = \int_{V_i}^{V_f} p dV$$

Question 1 / Vraag

Calculate the following to **3 significant figures**:

Bereken die volgende tot **3 beduidende syfers**:

[7]

1.1	$\left(10 \times \left(\frac{1.08 \times 10^4}{300} \right)^{\frac{1}{2}} + 4 \right)^{\frac{1}{2}} =$	8.00
1.2	$\ln \left(\frac{1}{e^2} \right) =$	-2.00
1.3	$\log(12.34^2) =$	2.18
1.4	$(\log 12.34)^2 =$	1.19
1.5	$\cos(-48^\circ) =$	0.669
1.6	$\tan^{-1}(-57.5)$ in radians / in radiale	-1.55 rad
1.7	Convert 5.88 radians to degrees Skakel 5.88 radiale na grade toe	337°

1.8 A circular race track has an internal circumference of 2 km and a width of 6 m. Calculate the area of the paving on the track in square metres.

'n Sirkelvormige resiesbaan het 'n binneomtrek van 2 km en 'n padwydte van 6 meter. Bereken die oppervlakte van die plaveisel van die pad in vierkante meter.

[5]

Determine the inner radius of the track:

$$c = 2\pi r$$

$$r_i = \frac{c}{2\pi} = \frac{2000 \text{ m}}{2\pi} = 318.31 \text{ m}$$

Outer radius:

$$r_o = r_i + 6 \text{ m} = 318.31 + 6 \text{ m} = 324 \text{ m}$$

Area of paving:

$$Paved\ Area = Total\ area - inner\ area$$

$$= \pi r_o^2 - \pi r_i^2$$

$$= \pi (r_o^2 - r_i^2)$$

$$= \pi ((324)^2 - (318)^2)$$

$$= 12101\ m^2$$

- 1.9 Iron has a density of $7.87\ g/cm^3$, and the mass of an iron atom is $9.27 \times 10^{-26}\ kg$. If the atoms are spherical and tightly packed, what is the volume of an iron atom?

Yster het 'n digtheid van $7.87\ g/cm^3$, en die massa van 'n yster-atoom is $9.27 \times 10^{-26}\ kg$. Indien die atome sferies is en styf gepak is, wat is die volume van 'n yster atoom? [5]

$$\text{Mass of iron atom: } 9.27 \times 10^{-26}\ kg = 9.27 \times 10^{-23}\ g$$

$$\text{Density} = \frac{\text{mass}}{\text{volume}} = \frac{n \times \text{mass of atom}}{\text{volume}}$$

$$n = \frac{\text{density} \times \text{volume}}{\text{mass of atom}} = \frac{7.87\ g/cm^3 \times 1\ cm^3}{9.27 \times 10^{-23}\ g}$$

$$= 8.49 \times 10^{22}\ \text{atoms}$$

Volume of 1 atom (assuming no space in between)

$$\text{volume of 1 atom} = \frac{\text{Volume with } n\ \text{atoms}}{n\ \text{atoms}}$$

$$= \frac{1\ cm^3}{8.49 \times 10^{22}\ \text{atoms}}$$

$$= 1.1779 \times 10^{-23}\ cm^3$$

$$= 1.18 \times 10^{-29}\ m^3$$

Question 2 / Vraag 2

- 2.1 Determine the first derivative of the function $y = (2x^2 - 3)^2$.

Bepaal die eerste afgeleide van die funksie $y = (2x^2 - 3)^2$. [2]

$$y = (2x^2 - 3)^2 = u^2$$

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

$$= 2(2x^2 - 3)(4x)$$

$$= 16x^3 - 24x$$

or

$$y = (2x^2 - 3)^2 = 4x^4 - 12x^2 + 9$$

$$\frac{dy}{dx} = 16x^3 - 24x$$

- 2.2 Determine the first derivative of the function. $y = \cos(2x^2 - 3)$.

Bepaal die eerste afgeleide van die funksie $y = \cos(2x^2 - 3)$. [3]

$$y = \cos(2x^2 - 3)$$

$$y = \cos(u)$$

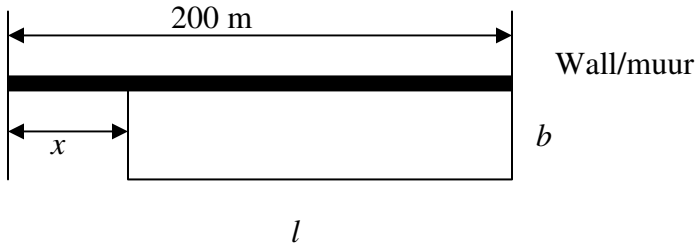
Then :

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} = [-\sin(2x^2 - 3)][4x] \\ &= -4x \sin(2x^2 - 3)\end{aligned}$$

- 2.3 An existing stone wall on a farm is about 200 m long. The farmer has enough material to erect a fence of length 100 m. He wishes to use a portion of the existing wall as the one side of a rectangular pen for his cattle and the other three sides are to be completed by using the available fencing material. Calculate the dimensions and the area of the pen which is constructed in this way and will have a maximum area. Remember to draw a sketch.

'n Lang reguit klipmuur wat 200 m lank is, is geleë op 'n plaas. Die boer besit genoeg kampmateriaal om 'n draadheining van 100 m lank te maak. Hy gebruik die bestaande klipmuur as een sy en voltooi die ander drie sye van 'n reghoekige kamp met sy beskikbare materiaal. Bereken die afmetings en oppervlakte van die kamp wat so gebou word dat die oppervlakte daarvan 'n maksimum sal wees. Onthou om 'n skets te teken.

[9]



$$A = l \times b$$

$$l = 200 - x$$

$$100 = 2b + l$$

$$\therefore b = \frac{1}{2}(100 - l)$$

$$\begin{aligned}A &= (200 - x)\left(\frac{1}{2}(100 - 200 + x)\right) \\ &= (200 - x)(-50 + \frac{1}{2}x) \\ &= -10000 + 150x - \frac{1}{2}x^2\end{aligned}$$

$$\frac{dA}{dx} = 0 \quad \text{for a max or min}$$

$$= \frac{d}{dx}[-10000 + 150x - \frac{1}{2}x^2]$$

$$= 150 - x$$

$$x = 150 \text{ m}$$

$$l = 200 - x = 50 \text{ m}$$

$$b = \frac{1}{2}(100 - l) = 25 \text{ m}$$

$$A = l \times b = 50 \times 25 = 1250 \text{ m}^2$$

- 2.4 (a) Calculate the area bounded by the t -axis and the graph of the function $v = 3 + 4t$, between $t = 1$ and $t = 5$. The units of v are meters per second.

Bereken die area tussen die t -as en die grafiek van die funksie $v = 3 + 4t$, tussen die waardes $t = 1$ en $t = 5$. Die eenhede vir v is meter per sekonde. [4]

$$\begin{aligned}
 v &= 3 + 4t \\
 \text{Area under curve} &= \int_1^5 (3 + 4t) dt \\
 &= \left[3t + \frac{1}{2}(4)t^2 \right]_1^5 \\
 &= (3(5) + 2(5)^2) - (3(1) + 2(1)^2) \\
 &= 60 \text{ m}
 \end{aligned}$$

- (b) What does this area in (a) represent and what are the units of this quantity?

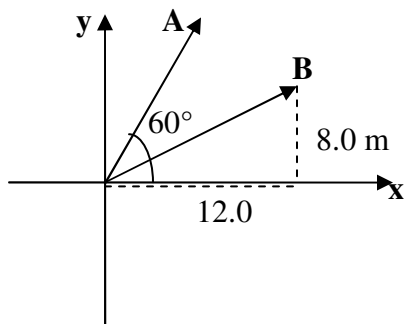
Wat word verteenwoordig deur die area in (a), en wat is die eenhede van hierdie hoeveelheid? [2]

Area represents the displacement
Units are in meters

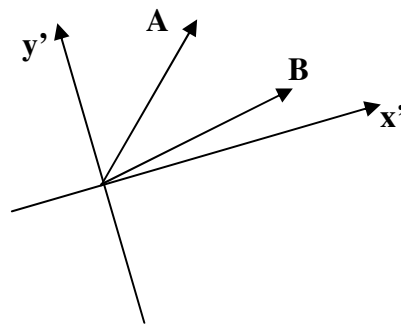
Question 3 / Vraag 3

- 3.1 $\vec{A}$ has the magnitude 12.0 m and is angled 60.0° counter-clockwise from the positive direction of the x axis of an xy coordinate system. Also, $\vec{B} = (12.0 \text{ m})\hat{i} + (8.00 \text{ m})\hat{j}$ on the same coordinate system. We now rotate the system counter-clockwise about the origin by 20.0° to form the $x'y'$ system. On this new system, what are $\vec{A}$ and $\vec{B}$ in unit-vector notation? Sketch the two situations.

$\vec{A}$ het 'n grootte van 12.0 m en wys in 'n rigting wat 'n hoek van 60.0° anti-kloksgewys met die positiewe x as maak in 'n xy koördinaatstelsel. $\vec{B} = (12.0 \text{ m})\hat{i} + (8.00 \text{ m})\hat{j}$ en is ook op dieselfde koördinaatstelsel. Ons roteer nou die koördinaatstelsel deur 'n hoek van 20.0° anti-kloksgewys om die oorsprong om die $x'y'$ stelsel te vorm. In hierdie nuwe koördinaatstelsel, wat is $\vec{A}$ en $\vec{B}$ in eenheidsvektor notasie? Skets die twee situasies. [2+10]



Original axes



Rotated axes

If we convert $\vec{B}$ to the magnitude-angle notation (as $\vec{A}$ already is) we have

$$|\vec{B}| = 14.4 \text{ m}$$

$$\theta_B = \tan^{-1}\left(\frac{8}{12}\right) = 33.7^\circ$$

After rotating: $|\vec{A}'| = 12.0 \text{ m}$ and $|\vec{B}'| = 14.4 \text{ m}$
 $\theta'_A = 60^\circ - 20^\circ = 40.0^\circ$ $\theta'_B = 33.7^\circ - 20.0^\circ = 13.7^\circ$

Converting these results to (x, y) representations, we obtain

$$A'_x = A \cos \theta' = 9.19 \text{ m}$$

(a) $A'_y = A \sin \theta' = 7.71 \text{ m}$

$$\vec{A}' = (9.19 \text{ m}) \hat{i}' + (7.71 \text{ m}) \hat{j}'$$

$$B'_x = B \cos \theta' = 13.99 = 14.0 \text{ m}$$

(b) $B'_y = B \sin \theta' = 3.4 \text{ m}$

$$\vec{B}' = (14.0 \text{ m}) \hat{i}' + (3.41 \text{ m}) \hat{j}'$$

3.2 Consider the following vectors:

Beskou die volgende vektore:

$$\vec{A} = (2.00 \text{ m}) \hat{i} + (3.00 \text{ m}) \hat{j}$$

$$\vec{B}: 4.00 \text{ m, at } +65.0^\circ$$

$$\vec{C} = (-4.00 \text{ m}) \hat{i} + (-6.00 \text{ m}) \hat{j}$$

$$\vec{D}: 5.00 \text{ m, at } -235^\circ$$

(a) Determine the sum of the four vectors in unit-vector notation.

Bepaal die som van hierdie vier vektore in eenheidsvektor-notasie.

[6]

Expressing $\vec{B} = (1.69\hat{i} + 3.63\hat{j}) \text{ m}$ and $\vec{D} = (-2.87\hat{i} + 4.10\hat{j}) \text{ m}$ in unit-vector notation, we have and, respectively.

(a) Adding corresponding components, we obtain the resultant

$$\begin{aligned} \vec{R} &= (A_x + B_x + C_x + D_x) \hat{i} + (A_y + B_y + C_y + D_y) \hat{j} \\ &= (-3.18 \text{ m}) \hat{i} + (4.73 \text{ m}) \hat{j} \end{aligned}$$

(b) Determine the magnitude and direction of the resulting vector in (a)

Bepaal die grootte en rigting van die resultante vector in (a).

[4]

The magnitude is

$$\begin{aligned} |\vec{R}| &= \sqrt{(R_x)^2 + (R_y)^2} \\ &= \sqrt{(-3.18 \text{ m})^2 + (4.72 \text{ m})^2} = 5.69 \text{ m.} \end{aligned}$$

The angle is

$$\begin{aligned}\theta &= \tan^{-1}\left(\frac{y}{x}\right) \\ &= \tan^{-1}\left(\frac{4.72 \text{ m}}{3.18 \text{ m}}\right) \\ &= 56.0^\circ \text{ above } -x \text{ axis (in 2nd quadrant)}\end{aligned}$$

3.3 Consider the following vectors $\vec{a} = 3.0\hat{i} + 3.0\hat{j} + 3.0\hat{k}$ and $\vec{b} = 2.0\hat{i} + 1.0\hat{j} + 3.0\hat{k}$.

Beskou die volgende vektore $\vec{a} = 3.0\hat{i} + 3.0\hat{j} + 3.0\hat{k}$ en $\vec{b} = 2.0\hat{i} + 1.0\hat{j} + 3.0\hat{k}$.

(a) Determine the angle between the two vectors.

Bepaal die hoek tussen die twee vektore.

[4]

$$|\vec{a}| = a = \sqrt{3^2 + 3^2 + 3^2} = \sqrt{27} = 5.2$$

$$|\vec{b}| = b = \sqrt{2^2 + 1^2 + 3^2} = \sqrt{14} = 3.7$$

$$\vec{a} \cdot \vec{b} = ab \cos \theta = a_x b_x + a_y b_y + a_z b_z$$

$$\cos \theta = \frac{a_x b_x + a_y b_y + a_z b_z}{ab}$$

$$= \frac{(3)(2) + (3)(1) + (3)(3)}{(5.2)(3.7)}$$

$$= 0.936$$

$$\theta = 20.7^\circ$$

This final answer depends on rounding. If it has been marked incorrect, and you have a final answer around 20° , you can bring it to me for re-marking.

(b) Determine a vector which is at right angles to both these vectors.

Bepaal 'n vektor wat 'n 90° hoek maak met al twee hierdie vektore.

[4]

$$\begin{aligned}\vec{a} \times \vec{b} &= (a_y b_z - b_y a_z)\hat{i} + (a_z b_x - b_z a_x)\hat{j} + (a_x b_y - b_x a_y)\hat{k} \\ &= ((3)(3) - (1)(3))\hat{i} + ((3)(2) - (3)(3))\hat{j} + ((3)(1) - (2)(3))\hat{k} \\ &= (6)\hat{i} + (-3)\hat{j} + (-3)\hat{k}\end{aligned}$$

Question 4 / Vraag 4

4.1 When the temperature of a copper coin is raised by 100°C , its diameter increases by 0.20%. To two significant figures, answer the following (given reasons for your answers):

Die diameter van 'n koper muntstuk verander met 0.20% wanneer die temperatuur met 100°C styg.

Antwoord die volgende, en gee jou antwoord tot 2 beduidende syfers (Gee ook redes vir jou antwoorde).

(a) Determine the percentage increase in the area of the face (top side) of the coin.

Bepaal die persentasie toename in die area van die bokant van die muntstuk.

[5]

$$\text{If } \Delta D = 0.20D_0, \text{ then } \Delta r = 0.20r_0$$

$$A = \pi r^2$$

First derivative gives the small change in A when there is a change in r

$$\frac{dA}{dr} = 2\pi r$$

$$dA = 2\pi r dr$$

$$\Delta A = 2\pi r \Delta r$$

$$\frac{\Delta A}{A} = \frac{2\pi r \Delta r}{\pi r^2} = \frac{2\Delta r}{r} = \frac{2(0.2r)}{r}$$

Therefore $\Delta A = 2\Delta r = 2(0.20) = 0.40\%$ of A

Do NOT use actual values to determine the change in area.

You can also get the same answer by using:

$$A = \pi r^2$$

$$A' = \pi (r + \Delta r)^2$$

$$\Delta A = A' - A$$

$$\% \text{ change in } A = \frac{\Delta A}{A} \times 100$$

- (b) Determine the percentage increase in the thickness of the coin.

Bepaal die persentasie toename in die dikte van die muntstuk.

[2]

Expansion is the same in all directions and therefore the increase in thickness is 0.20%

- (c) Calculate the coefficient of linear expansion of the coin.

Bereken die linieêre uitsettings-koëffisiënt van die muntstuk.

[3]

$$\Delta D = D_0 \alpha \Delta T$$

$$\alpha = \frac{\Delta D}{D_0 \Delta T} = \frac{(0.20 \times 10^{-2}) D_0}{D_0 \times (100^\circ \text{C})} = 2.0 \times 10^{-5} / ^\circ \text{C}$$

- 4.2 What mass of steam at 100°C must be mixed with 150 g of ice at its melting point, in a thermally insulated container, to produce liquid water at 50.0°C ?

Water massa stoom by 100°C moet met 150 g ys by dié se smelt punt gemeng word in 'n termies geïsoleerde houer, om vloeistof water by 50.0°C te kry?

[6]

$$c_{\text{water}} = 4180 \text{ J/kgK}$$

$$L_{f,\text{water}} = 333 \text{ kJ/kg}$$

$$L_{v,\text{water}} = 2256 \text{ kJ/kg}$$

Let the mass of the steam be m_s and that of the ice be m_i . Then

$$\Delta Q = 0 = Q_{\text{ice/water}} + Q_{\text{steam/water}}$$

$$Q_{\text{ice/water}} = -Q_{\text{steam/water}},$$

$$L_F m_c + c_w m_c (T_f - 0.0^\circ \text{C}) = m_s L_s - m_s c_w (T_f - 100^\circ \text{C})$$

where $T_f = 60^\circ \text{C}$ is the final temperature. We solve for m_s :

$$L_F m_c + c_w m_c (T_f - T_{i,ice}) = m_s L_s - m_s c_w (T_f - T_{i,steam})$$

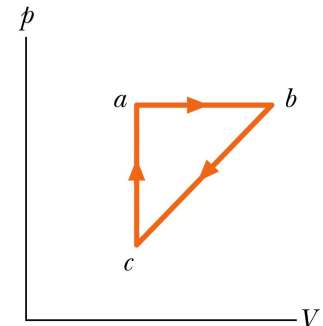
$$m_s = \frac{L_F m_c + c_w m_c (T_f - 0.0^\circ\text{C})}{L_s - c_w (T_f - 100^\circ\text{C})}$$

$$= \frac{(333 \times 10^3 \text{ kJ/kg})(150 \times 10^{-3} \text{ kg}) + (4180 \text{ J/kg.K})(150 \times 10^{-3} \text{ kg})(50^\circ\text{C} - 0.0^\circ\text{C})}{(2256 \times 10^3 \text{ kJ/kg}) - (4180 \text{ J/kg.K})(50^\circ\text{C} - 100^\circ\text{C})}$$

$$= 32.98 \times 10^{-3} \text{ kg} \approx 33.0 \text{ g}$$

4.3 A lab sample of gas is taken through a cycle $abca$ shown in the p - V diagram. The net work done is $+1.5 \text{ J}$. Along path ab , the change in the internal energy is $+3.0 \text{ J}$ and the magnitude of the work done is 5.0 J . Along path ca , the energy transferred to the gas as heat is $+2.5 \text{ J}$.

'n Laboratorium gasmonster word deur die siklus $abca$ geneem, soos aangedui in die p - V diagram. Die netto arbeid verrig is $+1.5 \text{ J}$. Langs pad ab , is die verandering in interne energie $+3.0 \text{ J}$ en die grootte van die arbeid verrig is 5.0 J . Langs pad ca word $+2.5 \text{ J}$ energie oorgedra na die gas.



(a) How much energy is transferred as heat along path ab ?

Hoeveel energie word oorgedra as warmte langs die pad ab ?

[3]

We note that process a to b is an expansion, so $W > 0$ for it. Thus, $W_{ab} = +5.0 \text{ J}$.

We are told that the change in internal energy during ab process is $+3.0 \text{ J}$, so application of the first law of thermodynamics for that ab yields

$$\Delta E_{ab,int} = Q_{ab} - W_{ab}$$

$$Q_{ab} = \Delta E_{int} + W$$

$$= +3.0 \text{ J} + 5.0 \text{ J}$$

$$Q_{ab} = +8.0 \text{ J}.$$